ASCII MASTER FLOW — PANEL 1/12: Karst coupled system

```
RECHARGE -> joints, bedding and sink points
DISSOLUTION -> openings enlarge underground
CONDUITS -> springs export water and solute
SURFACE + CAVE -> one hydrological landscape
MUST REMEMBER: Karst requires soluble rock, joints, water and carbonic-acid solution, followed
  by underground drainage and carbonate deposition; lapies, sinkholes, dolines, uvalas,
  poljes, caves, stalactites and stalagmites form a connected hydrological system.
```

ASCII MASTER FLOW — PANEL 2/12: Carbonate cycle

```
CO2 + WATER -> weak carbonic acid
ACID + CaCO3 -> dissolved bicarbonate
TRANSPORT -> solution moves through conduits
DEGASSING -> calcite can precipitate
```

ASCII MASTER FLOW — PANEL 3/12: Karst prerequisites

```
ROCK -> soluble, thick and exposed to recharge
STRUCTURE -> joints and bedding pathways
FLOW -> gradient plus outlet / base level
TIME -> sustained dissolution and removal
```

ASCII MASTER FLOW — PANEL 4/12: Hydrological zones

EPKARST -> shallow storage and routing
VADOSE -> descending water in air-filled voids
WATER TABLE -> fluctuating boundary
PHREATIC -> saturated tubes and loops

ASCII MASTER FLOW — PANEL 5/12: Surface-karst ladder

```
KARREN -> small solution grooves
DOLINE -> closed depression
UVALA / POLJE -> larger composite / structural basin
SWALLOW HOLE -> surface stream transfers underground
```

ASCII MASTER FLOW — PANEL 6/12: Cave-development sequence

```
INCEPTION -> favourable fracture or bedding plane
PHREATIC -> passage enlarges under saturation
VADOSE -> stream incises after water-table fall
LATE CHANGE -> collapse, fill or renewed solution
```

ASCII MASTER FLOW — PANEL 7/12: Speleothem map

STALACTITE -> roof drip deposit
STALAGMITE -> floor splash / drip deposit
COLUMN -> roof and floor forms join
FLOWSTONE / CURTAIN -> film flow on wall or slope

ASCII MASTER FLOW — PANEL 8/12: India cave firewall

```
MAWMLUH / MAWSMAI / SIJU -> Meghalaya limestone
BORRA / BELUM -> natural caves in Andhra Pradesh
AJANTA / ELLORA -> human rock-cut architecture
RULE -> location does not replace genetic classification
```

ASCII MASTER FLOW — PANEL 9/12: Proxy pipeline

```
CLIMATE -> rainfall, vegetation and soil CO2
RECHARGE -> route and residence time
CALCITE -> isotope / element signal
CHRONOLOGY -> dated series with uncertainty
```

ASCII MASTER FLOW — PANEL 10/12: Meghalayan hierarchy

```
QUATERNARY -> Period
HOLOCENE -> Epoch
MEGHALAYAN -> Age / Stage
GSSP -> KM-A speleothem at Mawmluh
CLOSE DISTINCTION: Solution enlarges voids whereas deposition builds speleothems; stalactites
  hang, stalagmites rise, columns join them, and Meghalaya caves, Bhimbetka shelters and the
  Meghalayan chronostratigraphic unit are categorically different.
```

ASCII MASTER FLOW — PANEL 11/12: Status and narrative caution

```
4.2 ka EVENT -> correlation guide
GSSP POINT -> physical lower boundary
MEGHALAYAN -> named formal interval
ANTHROPOCENE -> not a ratified Epoch
EVIDENCE LIMIT: Cave age, host-rock age, occupation age and formal geological-age boundary
  must not be collapsed; tourism and infrastructure effects depend on recharge, passage
  geometry, biodiversity and carrying capacity.
```

ASCII MASTER FLOW — PANEL 12/12: Karst conservation spine

```
MAP -> recharge, passages, springs and values
AVOID -> irreversible quarry and pollution pathways
MANAGE -> visitation, waste and water use
MONITOR -> hydrology, biodiversity and structural change
```